



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *Smart Hospital Assistant Robot for Autonomous Medicine Delivery*
by

Group: 08
Section: 09

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Abstract

The aim of this project is to design and develop a Smart Hospital Assistant Robot for Autonomous Medicine Delivery which will reduce the burden of healthcare workers and improve the efficiency, accuracy and safety of the common medication delivery. The proposed system uses ESP32 based robotic platform coupled with line following sensor, obstacle detection using ultrasonic sensor, ESP32-CAM and the control and verification – computer system with the servo-controlled medicine compartment. Under consideration is the robot's ability to autonomously navigate from a medicine storage station to the assigned patient beds, planning the route on pre-programmed routes and navigating obstacles within the path. Camera based identification can also verify patient information prior to unlocking the medicine compartment, avoiding delivery of unauthorized medication. At the end of the delivery, the robot will automatically go back to the medicine station and enter the standby mode for the next delivery. The prototype shows the integration of embedded systems, robotics, sensors, wireless communication and automation solutions to help facilitate repetitive health care operations. Overall, it is a standalone robot-assisted drug delivery system to support the delivery of medicine in a practical and affordable way.

Keywords

Autonomous Robot, ESP32, Medicine Delivery, Line Following Robot, Embedded Systems

1. Introduction

Problem and Motivation: The timely and accurate delivery of medicines to patients is essential for hospitals and other health care facilities. Healthcare professionals often have to make multiple trips between medicine storage spaces and patient beds to pick up and distribute prescribed medication. This is a required task, but also takes up a lot of staff time and can lead to delivery mistakes, delays and unnecessary physical contact. As robotics and embedded systems become more common, there is an opportunity to automate these repetitive tasks and enable the healthcare professional to spend more time on patient care.

Primary Objective: The main objective of this project is to design and develop a Smart Hospital Assistant Robot for Autonomous Medicine Delivery. The system is designed to be able to deliver medicines automatically in an autonomous way from a storage station to a selected medicine bed with navigation, obstacle detection, medicine security and patient verification. The proposed robot includes an ESP32 as its main controller, an ESP32-CAM for visual identification, IR sensors for line following navigation, an ultrasonic sensor for obstacle detection and compartments controlled by servos for secure medicine transportation. Task assignment and information processing is done on a PC-based system.

Scope and Relevance: The project's goal is to build a working prototype in a hospital simulation that has multiple patient beds, a medicine storage station and predetermined navigation routes. The robot is supposed to be programmed to receive a delivery order, travel to the specified delivery locations, securely transport the medication, verify the destination, deliver the medication and then automatically return to the base station. Because it demonstrates how low-cost robotics, wireless communication, sensors and automation can be used to increase the effectiveness and dependability of routine hospital tasks, the project is relevant to smart healthcare systems.

2. Related Work/ Inspiration

Related Work and Project Inspiration: This project was inspired by past studies of autonomous medicine delivery robots. Siyum et al. (2025) designed a smart line-following robot which is used to deliver medicines in the hospitals by using IR sensors, ultrasonic sensors, servo motors and notification system. Their project proved that simple sensors and robotics can be used to automate mundane medicine delivery in a healthcare setting. Likewise, Kavirayani et al. (2020) suggested a medicine delivery robot that utilizes sensors and indicators in the hospital to move towards patients. They were all about minimizing direct patient-contact with healthcare workers and ensuring timely medicine delivery. These studies inspired us to investigate the possibility of combining autonomous navigation, embedded systems, sensors and wireless communication into a viable system for delivering medicine in a hospital.

Project Differentiation and Problem Addressed: Our Smart Hospital Assistant Robot for Autonomous Medicine Delivery takes these methods further by not only delivering medicine autonomously but also ensuring its security and accurate delivery. The proposed system features line-following navigation, obstacle detection, ESP32 and ESP32-CAM, Wi-Fi communication with a PC, camera medicine and patient/destination verification, and a medicine compartment controlled by a servo motor, which is locked during transportation. The robot also has the ability to complete an end-to-end workflow by moving to the medicine station, safely picking-up and carrying the medicine, checking the destination before releasing it, and automatically returning to the medicine station afterwards. These features are designed to overcome problems like delivery of medicines to the wrong address, unauthorized entry of medicines during transit and manual intervention after every delivery.

3. Technical Approach

System Architecture: The main controller of the Smart Hospital Assistant Robot is based on the ESP32, which is used to integrate navigation, obstacle detection, medicine delivery, and communication. The robot is controlled by IR sensors which move the robot along a preprogrammed path, and an ultrasonic sensor detects the obstacles and halts the robot until it is clear of the obstacles. The PC uses Wi-Fi to assign tasks, monitor the system and verify with the patient/destination, while the ESP32-CAM captures the ID information to verify. The ESP32 is used to control the DC gear motors using the L298N motor driver, and to control the servo-controlled medicine compartment, while the LCD and buzzer give status indication. The robot receives a delivery command and then moves the medicine to the selected bed, checks the destination, releases the medicine if it is successful, and then returns to the medicine station and standby mode.

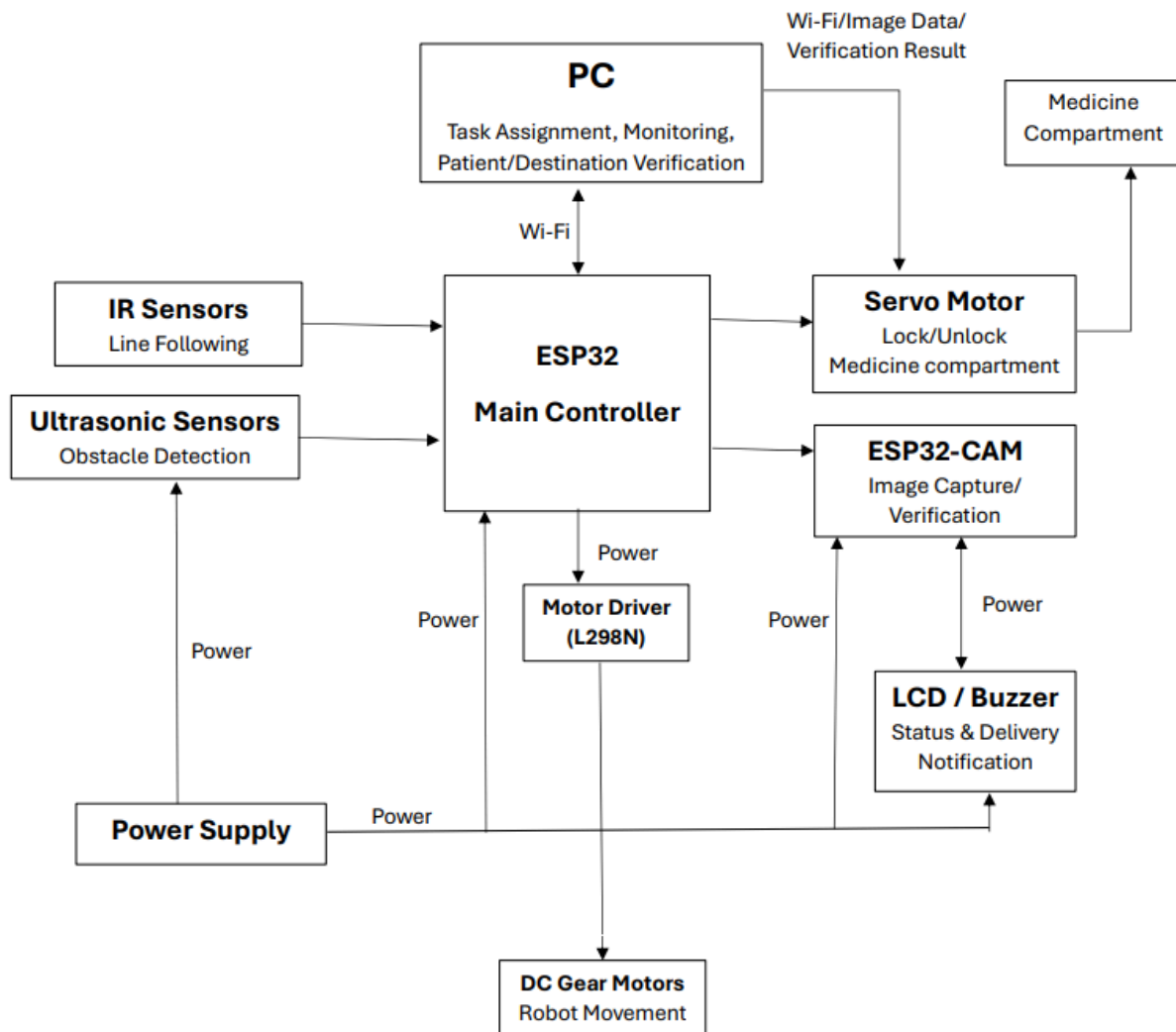


Figure: Block Diagram

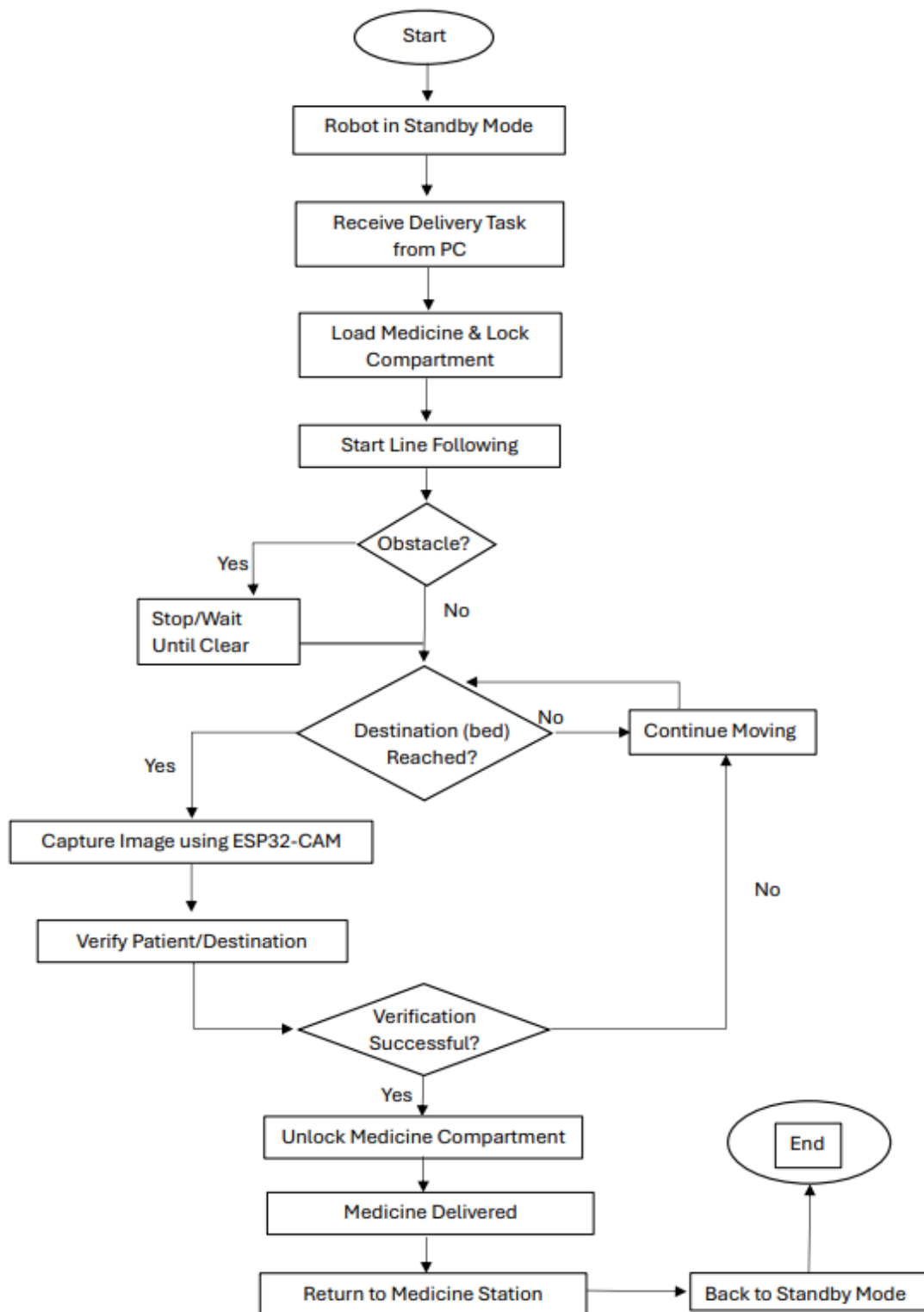


Figure: Flowchart

Components Used:

Hardware Components

Component	Quantity	Brief Description
ESP32	1	Main microcontroller used to control the robot and communicate with other components.
ESP32 Cam	1	Provides real-time visual monitoring for the robot.
2WD Chassis	1	Provides the robot with two-wheel movement.
L298 Motor Driver	1	Controls the speed and direction of the DC motors.
SG90 Servo Motor	2	Used for controlled mechanical movement, such as opening/closing the medicine compartment.
Breadboard	1	Used for temporary circuit connections during development.
Jumper Cables	—	Used to connect electronic components and modules.
Battery	1	Provides power to the robot's electronic and mechanical components.
Jumper Box	1	Provides organized jumper-wire connections.

Ultrasonic Sensor	1	Detects obstacles by measuring the distance in front of the robot.
5-Array Sensor	1	Used for line detection and tracking.
I2C Display	1	Displays information such as robot status or delivery messages.
L3259B	2	Used as part of the electronic control and interfacing circuitry.
F-F Cable	1	Provides female-to-female electrical connections between modules.
ESP32 T-C	1	Used for programming and development of the ESP32 system.
CP2102 TTL	1	Provides USB-to-TTL serial communication for programming/debugging.
LN2596	1	Used for voltage regulation in the power supply circuit.
Resistors	—	Used for current limiting and voltage control in the circuit.
On/Off Switch	1	Controls the main power supply of the robot.
8-Channel Logic Level Converter	1	Converts voltage levels between electronic components operating at different logic levels.

1N4148 Diodes	5	Used for current protection and signal control.
470 μ F Capacitors	2	Helps stabilize the power supply and reduce voltage fluctuations.
5V Buzzer	1	Provides an audible indication when an event occurs, such as medicine delivery.

Software Tools

Software	Purpose
Arduino IDE	Used to write, compile, upload and debug the program for the ESP32-based robot.

Cost Breakdown:

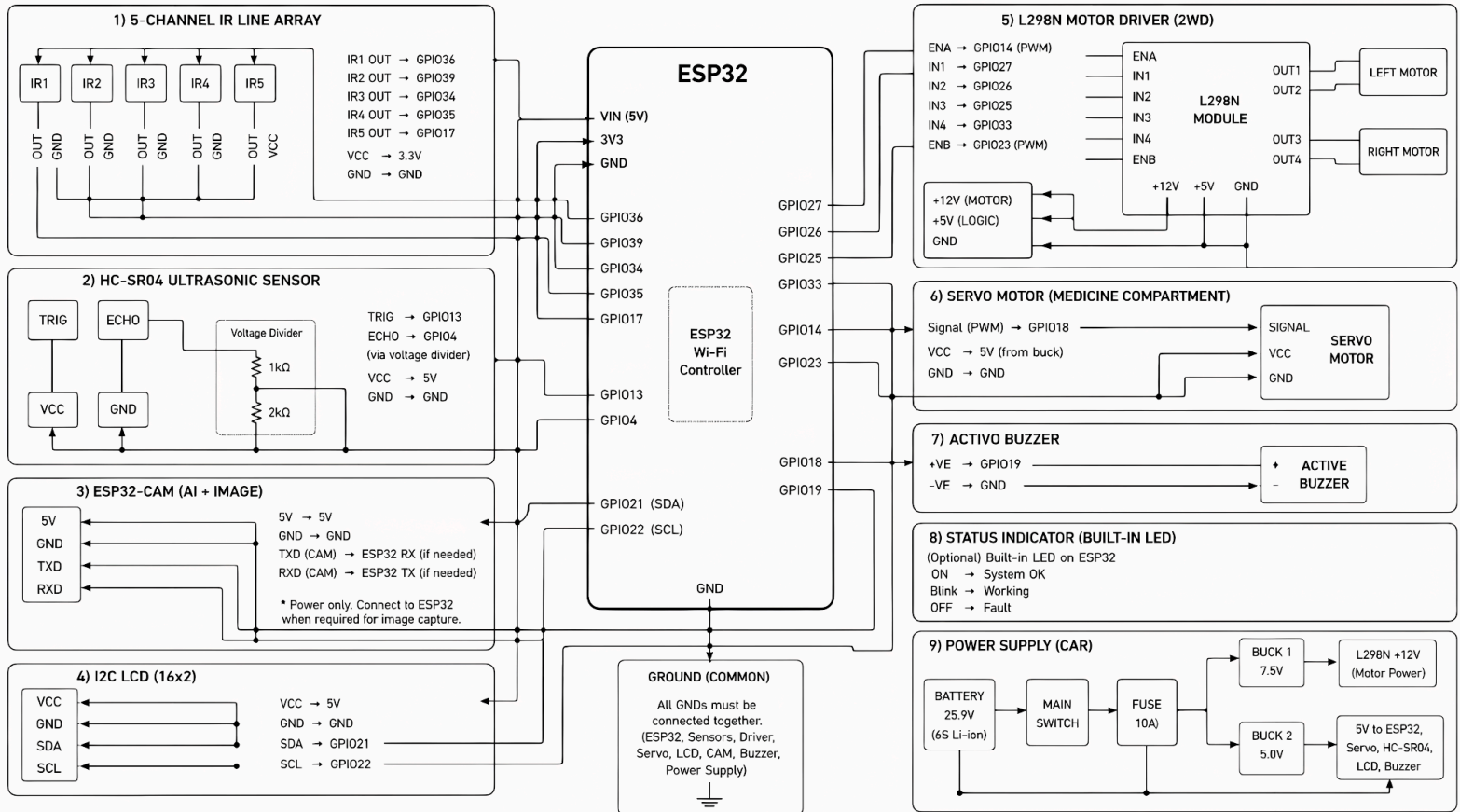
No.	Component	Qty.	Cost (BDT)
1	ESP32	1	400
2	ESP32 Cam	1	650
3	2WD	1	400
4	L298 Motor Driver	1	120
5	SG90 Servo Motor	2	200
6	Medium Breadboard	1	60
7	M-M & M-F Cables	2	130
8	1.1V 500 mAh Battery	1	650
9	Jumper Box	1	120
10	Ultrasonic Sensor	1	65
11	5-Array Sensor	1	300

12	I2C Display	1	300
13	L3259B	2	120
14	F-F Cable	1	65
15	ESP32 T-C	1	400
16	CP2102 TTL	1	200
17	LN2596	1	60
18	Resistors (200Ω, 2.2kΩ, 1kΩ)	150	30
19	On/Off Switch	1	5
20	8-Channel Logic Level Converter	1	120
21	1N4148 Diode	5	5
22	470 μF Capacitor	2	10
23	5V Buzzer	1	10

	Total Estimated Cost		₹4,840
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Circuit/Schematic:

SMART HOSPITAL ASSISTANT ROBOT – WIRING DIAGRAM



Functionality: The Smart Hospital Assistant Robot can be used for autonomous medicine delivery from a medicine storage station to a given location within the service area of the hospital. Navigation, obstacle detection, wireless communication, camera-based verification and medicine cabinet with servomotor control are integrated in the system. The whole system is described below:

1. Standby Mode: The robot is first stationed at the medicine station in standby mode, waiting for a new delivery task.
2. Delivery Task Assignment: The delivery request is set up with the PC based control system. The task specifies the destination/patient Point where the medicine should be supplied.
3. Medicine loading & compartment locking: The necessary medicine is installed in the medicine compartment. This compartment is then closed with a servodriven locking mechanism, which is contained to stop the compartment opening up if it was to be stolen during transit. The compartment is then locked up with the servodriven locking system, which ensures that the compartment remains closed when it is taken on board a vehicle for transport.
4. Autonomous Navigation: Once command "deliver" is received, the ESP32 begins to move the robot. The IR sensors continuously scan on the predefined line on the floor and give input to ESP32 for the line-following.
5. Obstacle Detection: As the robot is moving, the ultrasonic sensor keeps monitoring if there is any obstacle in front of the robot. The robot halts when an obstacle is sensed and does not move until there is no blockage. After the obstacle is cleared, the robot moves towards the given destination.
6. Destination Detection: The robot will follow its preprogrammed path, until it arrives at the chosen patient or destination location.
7. Image Capture: Once the destination has been reached, the ESP32-CAM will take a picture of the destination, which will be used for destination or patient verification.
8. Patient/Destination Verification: The processed image or information for identification is inputted into a PC based image verification system. This helps to guarantee delivery of medication to the proper patient or destination.
9. Medicine Compartment Unlocking: Successful verification will lead the ESP32 to send the unlock command to the medicine compartment and the medicine will be collected using the servo motor.
10. Failed Verification Handling: If verification is not successful, the medicine compartment stays closed and the moving forward can be repeated. This minimises the risk of medicine delivery errors and unauthorised medicines.
11. Medicine Delivery Completion: Once the verified patient or authorized person picks-up the medicine, the delivery is deemed complete.
12. Go back to Medicine Station: The robot can return to the prescribed path to a medicine storage site.
13. Return to Standby Mode: Once the robot arrives at the medicine station, it goes back into standby mode and waits for the next medicine delivery command.

Challenges:

1. We wanted to use a bug converter, but we couldn't adjust the voltage of the bug converter.
2. We used a camera to scan the QR code. When we first ran the code, it worked but after integrating, it showed a problem, that is our computer was not getting the code and driver to run the program of the bot.
3. Due to the height of the IR, the value of PID showed some random messed up values.

4. Sustainability & Impact

Sustainability: The Smart Hospital Assistant Robot is a system which can carry out repeated medicine delivery tasks without the need of disposable parts for every task. Reducing the need for single-use batteries and enabling the robot to run on rechargeable battery packs. The energy consumption can also be reduced by keeping the ESP32, sensors, display and other components in low power or standby mode during the robot's non-delivery period. The prototype is built using modular electronic components like the ESP32, sensors, motors, and motor drivers, which allows for the replacement of any damaged parts without having to rebuild the system. Other structural components, such as the chassis, medicine compartment, wiring, and more, can also be reused or modified for future versions. The body and medicine compartment can be made with recyclable materials like acrylic, appropriate plastics or light metal, if possible. Electronic waste and battery disposal are still environmental issues, however, so damaged batteries and electronic parts should be disposed of in a way that is both appropriate and follows e-waste management procedures.

Impact: With potential operational and social advantages, the suggested system can be implemented in clinics, hospitals, assisted living facilities, etc. For routine medication transportation, healthcare personnel must constantly move between patient locations and medication storage areas. A portion of this can be automated, which can help cut down on monotonous work and free up staff time for other crucial duties involving direct patient contact. The secure medicine compartment and patient/destination verification mechanism aim to minimise the chances of unauthorised access and incorrect medicine delivery. Within controlled hospital settings, autonomous navigation and obstacle detection can also aid in safe transportation. Industrially, the project is an example of the potential of the use of relatively low-cost embedded systems, sensors, wireless communication and robotics in a healthcare automation system. These systems could be particularly beneficial in cases where the purchase of more advanced commercial hospital robots is not economically viable.

Future Work: The future development will add to the autonomy, reliability and applicability of the robot in larger health care settings. The current line following navigation system may be replaced by more advanced autonomous navigation technologies like LiDAR, SLAM, depth cameras or indoor mapping, which will enable the robot to move without the need for floor lines. Better computer vision would allow for quicker identification of patients and medicine on board without needing to heavily depend on an external PC. A dedicated hospital management application can also be developed to assign deliveries, keep track of delivery records, monitor the status of the robots, and manage multiple robots at once. Other sensors could enhance collision avoidance and environmental awareness. Future versions may incorporate

automatic charging or docking to enable the robot to recharge itself when the battery level is low. The medicine compartment may also be divided into several separate compartments, so that several medicines can be dispensed in one trip. These enhancements would enable the system to be scaled to a more powerful platform for use throughout several hospital rooms, wards or floors.

Limitations: Although the prototype was successfully implemented, there are still some limitations. IR line following sensors can only be used to follow a specific path and require a well-marked path for the robot to follow. Navigation may be inaccurate with changes in lighting, floor surfaces, or damaged route markings. The HC-SR04 ultrasonic sensor has a relatively basic obstacle detection function, and may fail to detect some objects or surfaces. The ESP32 and ESP32-CAM also have less processing power and memory than dedicated computers, limiting the advanced computer-vision and autonomous-navigation functions.

The system also requires the stable Wi-Fi communication between the robot and PC for task management and verification. System operation may be impacted by interruption of the network, weak signal strength, or communication delays. The lighting, camera angle, image quality, or scratches on the ID codes can affect camera-based verification. The robot's operating time is limited by the battery capacity and motors and servo motors may generate relatively high current demands. In addition, the prototype has been built for a controlled simulated hospital setting and has not been designed to cope with the presence of multiple moving obstacles, unrestricted human movement, or crowded corridors. Thus, further hardware, software, safety testing and system validation would be required before the robot could be used in a real healthcare institution.

5. Results & Discussion

The performance of the Smart Hospital Assistant Robot can be evaluated based on its major functions, including line-following navigation, obstacle detection, patient/destination verification, medicine compartment operation, return-to-base operation, and overall medicine delivery. Table 1 presents estimated prototype performance values for demonstration purposes based on 10 trials for each function.

Function Tested	Total Trials	Successful Trials	Success Rate (%)
Line-following navigation	10	9	90%
Obstacle detection and stopping	10	10	100%
Patient/destination verification	10	9	90%
Compartment lock/unlock	10	10	100%
Return to medicine station	10	8	80%
Complete medicine delivery	10	8	80%

The estimated results suggest that the individual subsystems would have generally reliable performance when tested on a prototype. The success rates of the obstacle detection and compartment locking and unlocking are the highest, estimated at 100%, indicating the expected reliability of the stopping mechanism and the servo-controlled medicine compartment. Line following navigation and patient/destination verification are estimated at 90%. IR sensor calibration, sharp turns, or track surface changes may cause occasional navigation failures, and verification performance may be impacted by camera position, lighting or image quality.

The estimated return to base and complete delivery success rates are reduced to 80% due to the fact that these functions rely on multiple subsystems functioning properly in the same trial. If a navigation or verification fails at any point, the entire delivery cycle may fail. However, the results estimated by the proposed system indicate that it can achieve the primary goals of the project: autonomous navigation,

obstacle detection, secure medicine transportation, destination verification, medicine release and automatic return to the medicine station. Additional calibration and physical testing would be needed to assess the actual performance and reliability of the entire system.

6. Conclusion

Project Recap and Main Findings: The Smart Hospital Assistant Robot was designed to automate medicine delivery in a hospital setting and enhance the efficiency, accuracy and security of medicine delivery. It features line following navigation, obstacle avoidance, wireless communication, camera patient/destination verification, and a servo controlled medicines cabinet. It is designed to be delivered by a PC, go to the assigned area, detect and avoid obstacles, verify the delivery area before unloading the medicine and automatically return to the medicine station. In conclusion, the project showcases the feasibility of embedding systems, sensors, communication and robotics into a system for delivering medicines autonomously.

Sustainability and Impact: The project takes a step towards sustainability by adopting rechargeable batteries, reusability of structural materials and modular electronic components which can be replaced individually without replacing the system as a whole. Automating the delivery of mundane medication allows healthcare workers to spend more time on patient care and reduces repetitive work. The system can also be used for the secure and controlled delivery of medicines, and be verified at the delivery point. Further development and scaling of the low-cost robotic system may prove to be a viable solution for healthcare automation in hospitals, clinics and nursing homes.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Plaban Basu	22101831	Software Development & Documentation	● Developed the control code for the medicine dispenser & lock box. ● Contributed to the project documentation.
Mahdi Hasan	22201760	Software Development & Assembly	● Implemented the code for ESP32 Cam module for QR-code verification.
Tamim Hasan Akib	22301115	Hardware Assembly & Testing	● Assembled the circuit of the robot. ● Tested and calibrated the robot.
Shami Al Bashar Prothoy	23101457	Software Development & Testing	● Programmed the code for the robot. ● Tested and calibrated the robot.
Atef Ashhab Rahman	24341287	Hardware Assembly & Documentation	● Assembled the medicine dispenser and lock box. ● Contributed to the project documentation

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